Notes from Ben, not part of the talk

- Side chain: now explained on slide 2 in one clause (Ben, 14 October). The whole deck was revised on 14 October along the four facts of slide 6, see `WORK-SPLIT.md`.

Highly accurate protein structure prediction with AlphaFold

Paper by Jumper et al., *Nature* 596 (2021)

Evita Bosch Pokunevich Ben Bullinger

Machine Learning for Genomics (263-5351-00L), ETH Zürich, Fall 2026

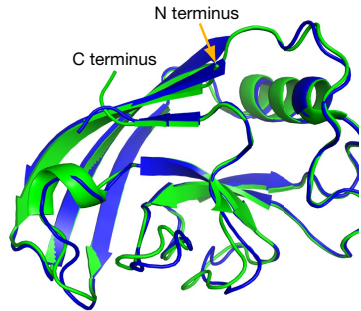
Outline

1. The problem
2. The benchmark
3. The network
   The data
   The network, in detail
4. Training
5. What makes it work
6. Conclusion
7. Questions

The problem: sequence to structure

SIFSYITESTGTPS…

sequence of amino acids. Each has the same backbone and its own side chain



3D coordinates of all heavy atoms
Prediction in blue, experiment in green

An experiment takes months to years

Billions of sequences are known

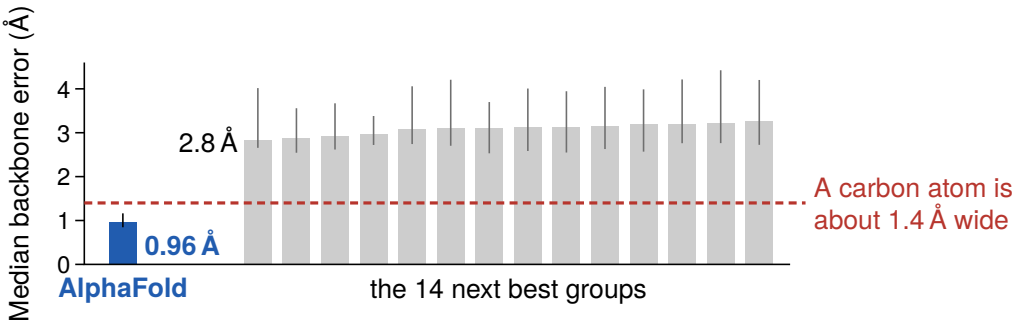
The problem is more than 50 years old

Sequence and structure are target T1049 of CASP14, the blind prediction contest, Protein Data Bank (PDB) entry 6Y4F, Fig. 1b.

The benchmark: CASP14

CASP14: Critical Assessment of protein Structure Prediction, May to July 2020

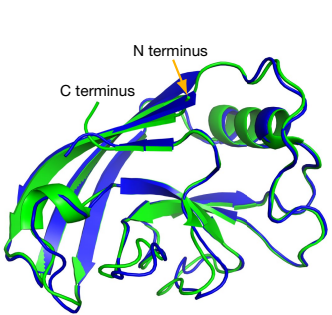
Blind test: experimental structures withheld until predictions were submitted



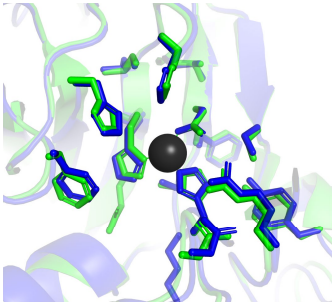
The median backbone error is below one atom width

Redrawn from Fig. 1a. Cα r.m.s.d.$_{95}$ on 87 domains (95% CIs). 1 Å = 10^{-10} m.

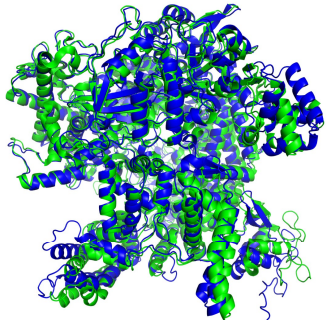
Prediction and experiment superposed



Backbone



Side chains

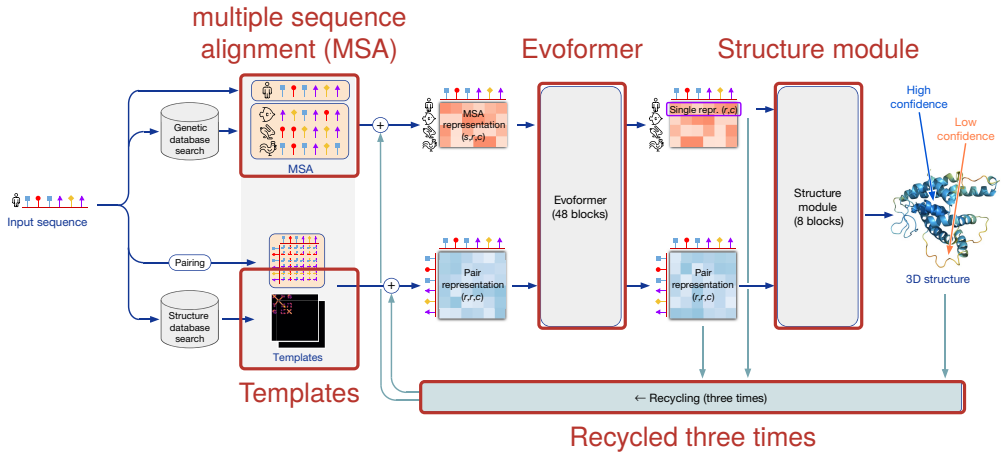


A 2,180-residue chain

AlphaFold experiment

Prediction and experiment overlap, also in the side chains

The pipeline: MSA in, coordinates out



Coordinates come out of the network itself

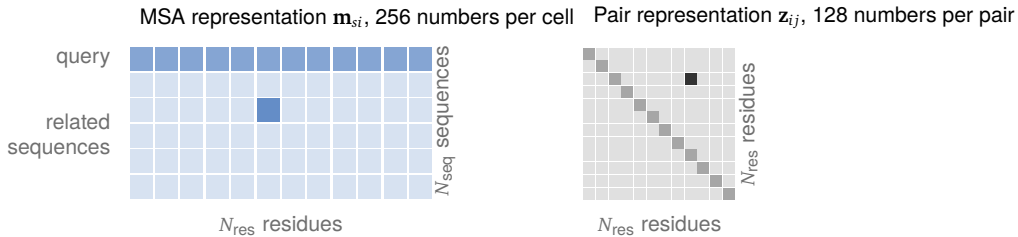
Four properties, four design choices

		Fact	Design choice in AlphaFold 2
1		An MSA is a set of sequences. A protein is a chain of residues.	Attention over the sequences themselves, along the rows and the columns.
2		Distances in 3D must close triangles.	The pairs are the edges of a residue graph, updated through triangles.
3		A structure is the same after any rotation or translation.	One frame per residue. Attention uses distances; updates in local frames.
4		A protein and its mirror image are different molecules.	The loss compares atoms in local frames. A mirror image is penalised.

Removing any one of the four costs accuracy. The ablations measure how much.

The two representations

Two tables of numbers, one per MSA cell and one per residue pair

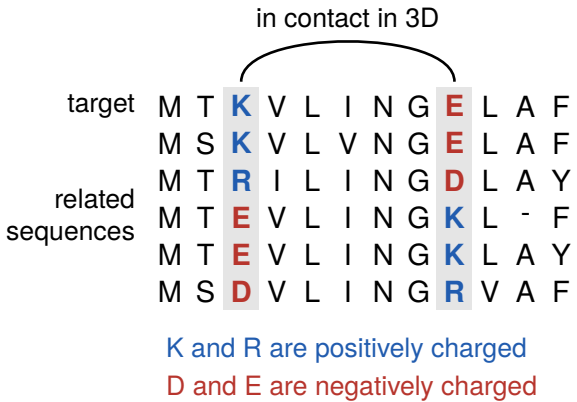


One vector per cell of the alignment, and one per ordered pair of residues

Everything the network learns is written into these two tables

Supplementary Information (SI) 1.5, Algorithms 3 and 4. Sizes in training: 128 sequences, crops of 256 residues (SI Table 4).

The input: MSA representation



A row is a related protein

A column is an aligned position

Contacts may change together

The network reads the sequences themselves

D. S. Marks et al., *PLoS ONE* 6, 2011. The alignment is an illustration, not data.

The input: Pair representation

What z_{ij} holds and where it comes from

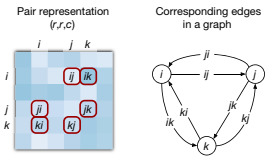


Fig. 3b, the pairs as directed edges

128 numbers for every ordered pair of residues, the edge $i \rightarrow j$ of the graph

Started from the two amino acids and their distance along the chain, clipped at 32. Templates and the previous cycle enter as distances.

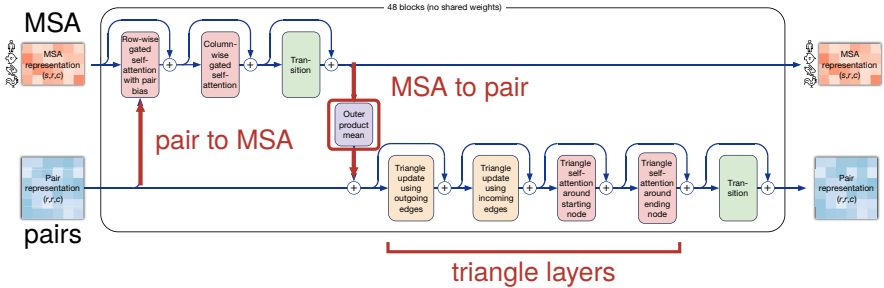
At the end, the distogram reads a distance distribution for every pair from it

Where the structure is held before there are coordinates

SI 1.5, Algorithm 4 (relative position), 1.7.1 (templates), 1.10 (recycling), 1.9.8 (distogram).

The Evoformer block

1. The sequences in an MSA are a set. The residues are a chain.

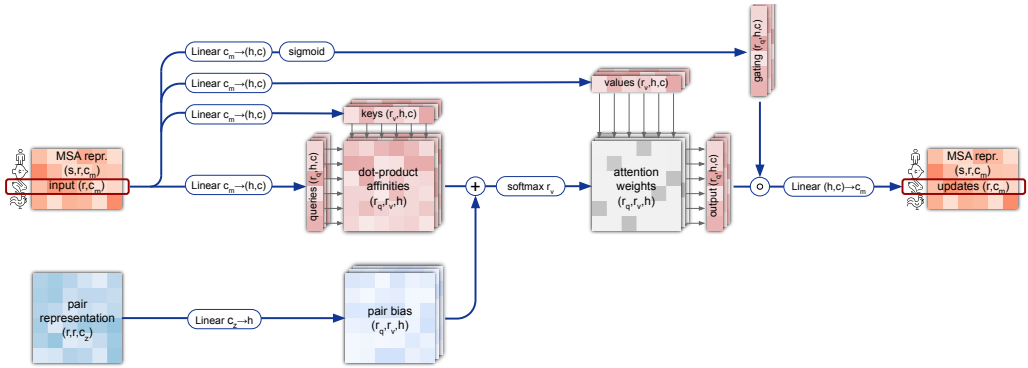


Attention along the rows and the columns. MSA and pair representation exchange information

The paper calls this graph inference. The edges are residues in proximity.

Evoformer: row attention with pair bias

Supplementary Fig. 2, Algorithm 7

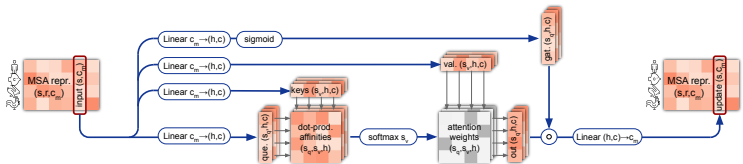


One sequence at a time: residue *i* attends to every residue *j* of the same sequence

The pair representation is added to the logits. The pairs steer the MSA.

Evoformer: columns and outer product mean

Supplementary Figs. 3 to 5, Algorithms 8 to 10



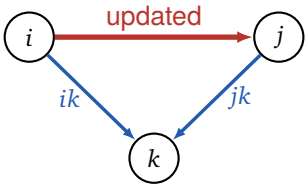
One residue at a time: the sequences attend to each other at position i, then a two-layer perceptron per cell

Outer product mean: the outer product of columns i and j, averaged over the sequences, is added to $\mathbf{z}_{ij}$

Here the alignment writes into the pairs, in every one of the 48 blocks

Evoformer: triangle updates

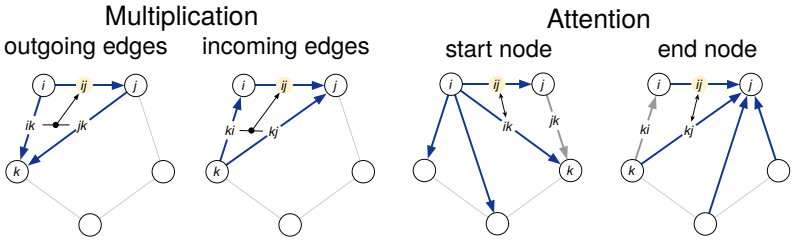
2. Distances in 3D must close triangles.



The pair representation is the edge set of the graph (Fig. 3b)

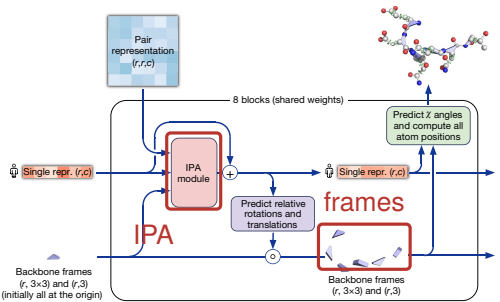
Edge ij is updated from the edges ik and jk

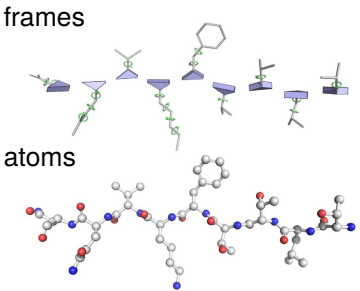
The triangle inequality acts as an inductive bias



Structure module: residues as rigid bodies

3. A structure is the same after any rotation or translation.





A frame is one rotation and one translation per residue

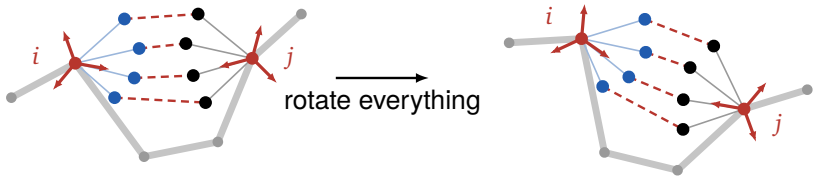
Invariant point attention (IPA) updates the residues, then all frames move at once

Nothing forces neighbouring residues to stay connected

Invariant point attention

3. A structure is the same after any rotation or translation.

$$a_{ij} = \text{softmax}_j\Big(\underbrace{\boldsymbol{q}_i^\top \boldsymbol{k}_j}_{\text{usual attention}} + \underbrace{b_{ij}}_{\text{pair bias}} - \underbrace{\textstyle\sum_p \left\| T_i \circ \vec{\boldsymbol{q}}_{ip} - T_j \circ \vec{\boldsymbol{k}}_{jp} \right\|^2}_{\text{distances between 3D points}} \Big)$$



Points are placed in the frames of i and j

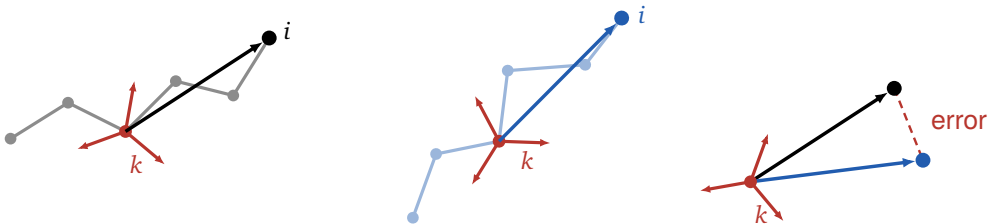
The distances stay the same

The weights are invariant to global rotation and translation

The formula is Algorithm 22 of the supplement for one head, without scaling constants.

Frame aligned point error (FAPE)

4. A protein and its mirror image are different molecules.



experiment prediction both, seen from frame k

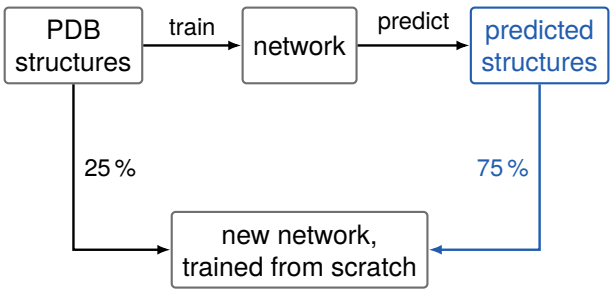
$$\text{FAPE} = \frac{1}{Z} \underset{k,i}{\text{mean}} \min\left(d_{\text{clamp}}, \left\|(\boldsymbol{R}_k, \boldsymbol{t}_k)^{-1}\boldsymbol{x}_i - (\boldsymbol{R}_k^{\text{true}}, \boldsymbol{t}_k^{\text{true}})^{-1}\boldsymbol{x}_i^{\text{true}}\right\|\right)$$

A rotation changes nothing. A mirror image is penalised.

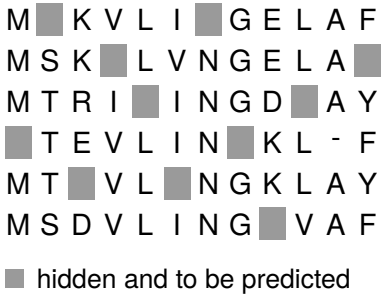
Supplementary Information, Algorithm 28, with $Z = d_{\text{clamp}} = 10\,\text{Å}$. The drawing is an illustration.

Training data: PDB and self-distillation

Self-distillation adds 350,000 structures



A masked-MSA loss is added

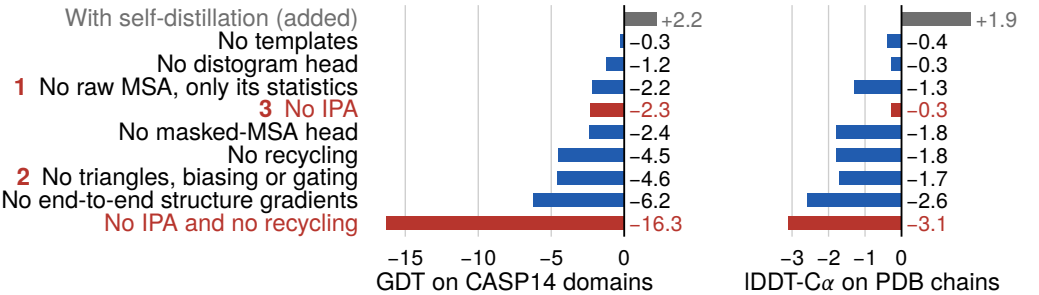


Training uses sequences without structures

Self-distillation follows Q. Xie et al., 2020. Masking follows BERT, J. Devlin et al., 2019.

Ablations: the cost of each component

1 2 3 4

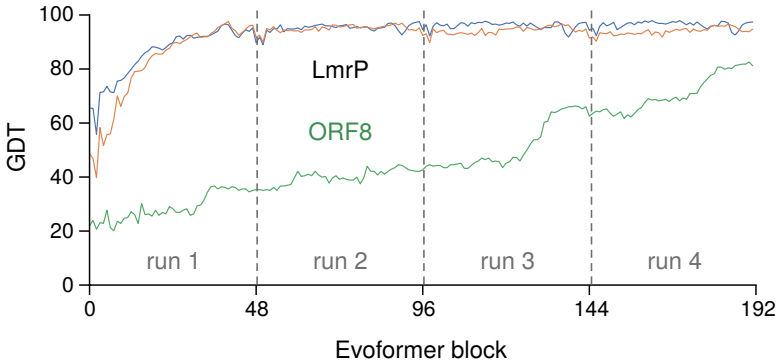


4 With a mirror-blind loss instead of FAPE, the model often builds the mirror image

Each of the four choices costs accuracy on CASP14 when it is removed

Redrawn from Fig. 4a, mean change vs baseline over training runs. GDT and lDDT 0–100. Row 3 (No IPA) removes iterative refinement but keeps frames and FAPE (SI 1.13.1). Line 4: Suppl. Fig. 9.

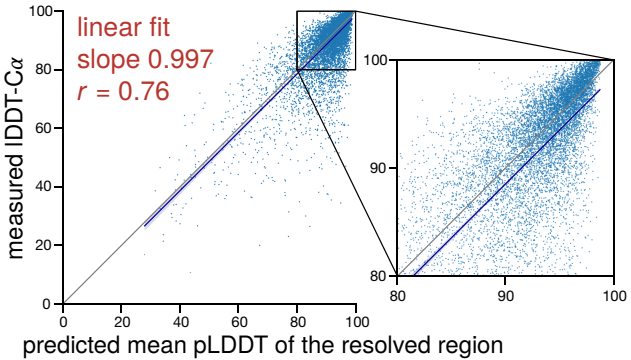
Intermediate structures in the trunk



The trunk is frozen and every block gets a structure module

Some proteins are solved early and others late

Confidence: pLDDT against measured lDDT

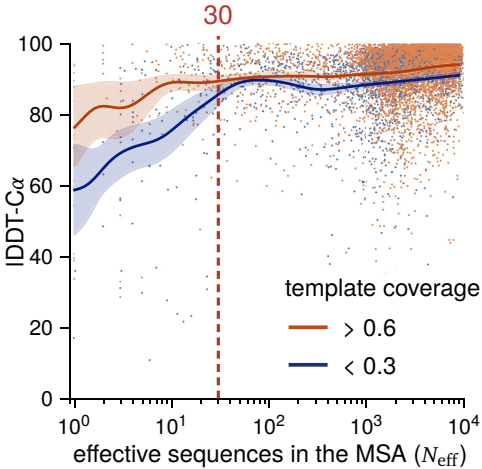


pLDDT is the predicted lDDT-Cα per residue

Predictions need a well-calibrated confidence

K. Tunyasuvunakool et al., *Nature* 596, 2021.

Limitations: MSA depth and complexes

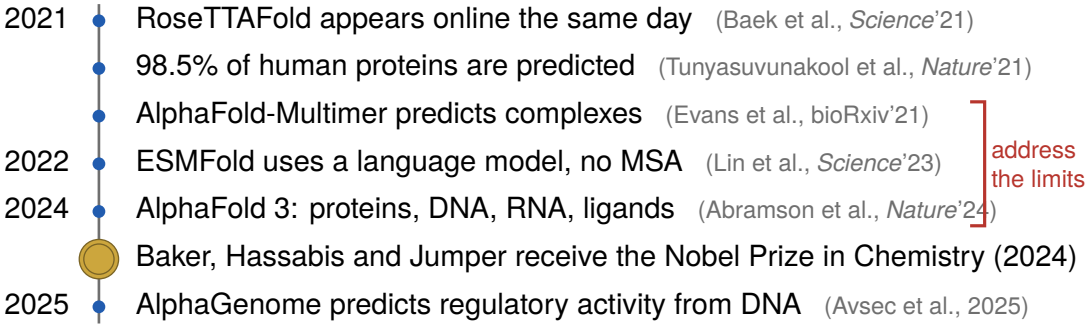


Accuracy falls below about 30 effective sequences

AlphaFold is weaker for chains shaped by other chains

AlphaFold needs enough related sequences

After AlphaFold 2: 2021 to 2025



AlphaFold 3 is more accurate than specialised tools, with one exception

Exception: AIchemy_RNA2 with human expert intervention on CASP15 RNA targets.

AlphaFold 3: which choices survived

Abramson et al., *Nature* 630 (2024), pages 493 to 497

1 2 3 4

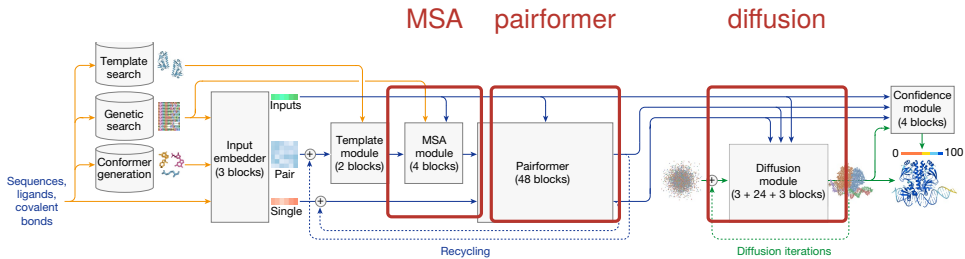
	AlphaFold 2	AlphaFold 3
1	attention over sequences	a small MSA block, and the MSA is not kept
2	triangles on the pairs	kept, 48 blocks
3	frames, IPA, equivariance	dropped for diffusion and random rotations
4	FAPE in local frames	dropped, 4.4 % chirality violations remain

The bitter lesson, with two footnotes. The graph survived. The hand came back.

Richard Sutton, *The Bitter Lesson*, 2019: “building in how we think we think does not work in the long run”.

The AlphaFold 3 architecture

Abramson et al., *Nature* 630 (2024), Fig. 1d



Tokens for proteins, nucleic acids, ligands and ions

Diffusion denoises random atom coordinates, 25 samples ranked by confidence

Less structure in the network, more in the data and in the sampling

AlphaFold 3: the cost of the removal

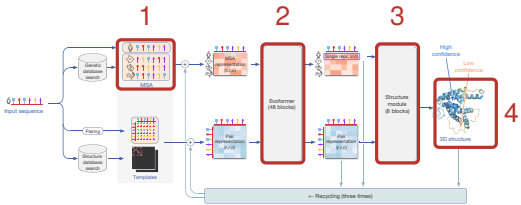
Abramson et al., *Nature* 630 (2024), pages 495 to 498

AlphaFold 2 had	AlphaFold 3 reports	The patch
the hand in the loss	4.4 % chirality violations	ranking, more seeds
a violation loss	clashes, overlapping chains	ranking
loops for disorder	hallucinated structure	AlphaFold-Multimer distillation
MSA attention	same MSA dependence	none

No ablation of the removed parts is reported

It wins everywhere but one RNA entry. The costs were paid in patches.

Summary: four design choices, measured



		GDT lost
1	attention over sequences	−2.2
2	triangles on the pairs	−4.6
3	frames and IPA	−2.3
4	FAPE in local frames	mirrored

Trained end to end. One prediction takes about a GPU minute

The median backbone error is below one atom width

J. Jumper et al., “Highly accurate protein structure prediction with AlphaFold,” *Nature* 596, 583–589 (2021). GDT lost on CASP14 when removed, Fig. 4a and Suppl. Fig. 9. Prediction time: 384 residues.

Exam question

Placeholder. The question from the course team arrives about seven days before the talk.

Our answer.

Questions for the audience

1 AlphaFold 2 built rotation equivariance and the hand into the network and the loss. AlphaFold 3 dropped both and learns them from data. When is it worth building a symmetry into a model, and when is it better learned?

2 Accuracy falls below about 30 related sequences. Has the network learned how proteins fold, or how evolution has varied them? What result would settle it?

3 pLDDT is trained on the model's own errors. On which proteins would you distrust it, and what should a genomics pipeline do with a low-confidence region?

Why is IPA invariant to rigid motions?

SI 1.8.2, Algorithm 22, equations 3 to 6



The only geometry in the weights is the distance between points that i and j place in their own frames

$$\left\|(T_{\text{global}} \circ T_i) \circ \vec{\boldsymbol{q}}_i - (T_{\text{global}} \circ T_j) \circ \vec{\boldsymbol{k}}_j\right\|^2$$
$$= \left\|T_{\text{global}} \circ (T_i \circ \vec{\boldsymbol{q}}_i - T_j \circ \vec{\boldsymbol{k}}_j)\right\|^2 = \left\|T_i \circ \vec{\boldsymbol{q}}_i - T_j \circ \vec{\boldsymbol{k}}_j\right\|^2$$

moving both points together leaves their distance unchanged

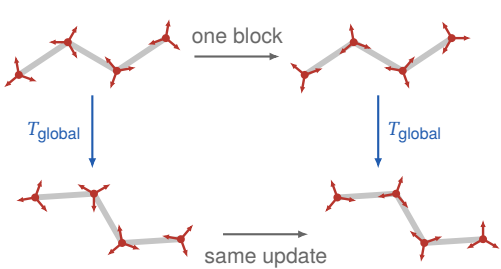
$$(T_{\text{global}} \circ T_i)^{-1} \circ \textstyle\sum_j a_{ij}\, (T_{\text{global}} \circ T_j) \circ \vec{\boldsymbol{v}}_j$$
$$= T_i^{-1} \circ T_{\text{global}}^{-1} \circ T_{\text{global}} \circ \textstyle\sum_j a_{ij}\, T_j \circ \vec{\boldsymbol{v}}_j = T_i^{-1} \circ \textstyle\sum_j a_{ij}\, T_j \circ \vec{\boldsymbol{v}}_j$$

the output points are read back in frame i, and T_{global} meets its inverse. Exact, not learned.

Weights and outputs do not depend on where the structure sits

Why is the structure module equivariant?

SI 1.8.2 and 1.8.3, Algorithms 20 and 23



The update of a frame is predicted from $\boldsymbol{s}_i$, which is invariant, and composed in the frame's own coordinates

$(T_{\text{global}} \circ T_i) \circ T_i^{\text{upd}} = T_{\text{global}} \circ (T_i \circ T_i^{\text{upd}})$

moving the input first or the output last gives the same frames

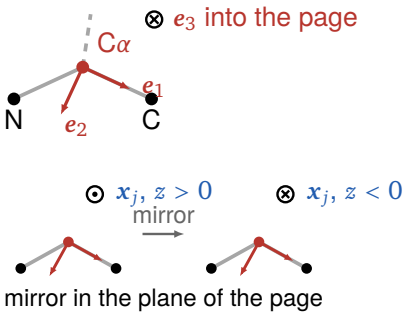
"Rigid motion" in the supplement means proper rotation and translation: SE(3). (Reflections are penalized by FAPE, backup 3).

All frames start as the identity, so the output lives in the initialisation frame; the loss must be invariant to it

Exact equivariance. Handedness is enforced by FAPE and chiral data.

Where does the handedness come from?

SI 1.8.1, 1.9.3 and 1.8.4, Algorithm 21, equations 11 to 17



A frame is built from N, Cα and C by Gram–Schmidt, with $\boldsymbol{e}_3 = \boldsymbol{e}_1 \times \boldsymbol{e}_2$. It is always right-handed.

Reflect all atoms. The frames do not reflect with them. In local coordinates exactly z flips:

$$T_i^{-1} \circ \boldsymbol{x}_j \;\mapsto\; \mathrm{diag}(1, 1, -1)\,(T_i^{-1} \circ \boldsymbol{x}_j)$$

$$\mathrm{FAPE}(X, X_{\mathrm{mirror}}) = 2 \sum_{ij} \left|(T_i^{-1} \circ \boldsymbol{x}_j)_z\right|$$

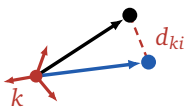
large unless the structure is flat

Trained with dRMSD, the error of the distance matrix, which is blind to reflection, the model builds the mirror image in a large share of cases (Suppl. Fig. 9)

The loss sees the hand, the data teach it. Side chains are always built L.

What does FAPE measure?

SI 1.9.2, 1.9.4 and 1.9.5, Algorithm 28, equations 18 to 30



seen from frame k

Every atom i in every frame k, in local coordinates: $N_{\text{frames}} \times N_{\text{atoms}}$ terms, each clamped at 10 Å and divided by 10 Å

Invariant to a rigid motion of the prediction, and of the truth, by the same cancellation as in IPA

Lower bound: FAPE is at least the RMSD after the best rigid alignment, whatever the frames (eqs. 18 to 22). So it is zero only if the two point sets differ by a proper rigid motion.

Non-negative, symmetric, triangle inequality: a pseudometric on structures (eqs. 23 to 30)

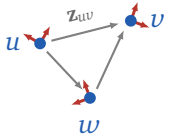
Local geometry in every frame: blind to placement, sensitive to the hand

Is AlphaFold 2 a graph neural network?

Formalism of Bronstein, Bruna, Cohen and Veličković, 2021

$$\mathbf{h}_u = \phi\left(\mathbf{x}_u, \bigoplus_{v \in \mathcal{N}_u} \psi(\mathbf{x}_u, \mathbf{x}_v, \mathbf{e}_{uv})\right)$$

message ψ, aggregation $\bigoplus$, update ϕ. Nodes: residues, $\mathbf{s}_u$ and frame T_u. Edges: $\mathbf{z}_{uv}$.



Block	message ψ	into
MSA row attention	$\psi(\mathbf{m}_{su}, \mathbf{m}_{sv}, \mathbf{z}_{uv})$, the edge biases the weight	node u
outer product mean	$\psi(\mathbf{m}_{\cdot u}, \mathbf{m}_{\cdot v})$, averaged over the sequences	edge uv
triangle updates	$\psi(\mathbf{z}_{uv}, \mathbf{z}_{uw}, \mathbf{z}_{vw})$ over third nodes w (out and in)	edge uv
IPA	$\psi(\mathbf{s}_u, \mathbf{s}_v, \mathbf{z}_{uv}, \lVert T_u \circ \vec{\mathbf{q}}_u - T_v \circ \vec{\mathbf{k}}_v \rVert)$, values moved by $T_u^{-1} \circ T_v$	node u
backbone update	none, ϕ alone	frame T_u

Triangles combine two edges into the third to make the graph consistent with 3D space.

Geometry enters through invariants: distances in weights, relative frames in values.

What does each block compute?

Supplement, Algorithms 6 to 32

	Update	Message	SI
Evoformer, 48 blocks			
1	$\mathbf{m} \leftarrow \mathbf{m} + \text{RowAttention}(\mathbf{m} \mid \mathbf{z})$	node → node, edge biases the weight	Alg. 7
2	$\mathbf{m} \leftarrow \mathbf{m} + \text{ColumnAttention}(\mathbf{m})$	across the sequences at one node	Alg. 8
3	$\mathbf{m} \leftarrow \mathbf{m} + \text{MLP}(\mathbf{m})$	per node	Alg. 9
4	$\mathbf{z} \leftarrow \mathbf{z} + \text{OuterProductMean}(\mathbf{m})$	node pair → edge	Alg. 10
5	$\mathbf{z} \leftarrow \mathbf{z} + \text{TriangleMultiplication}(\mathbf{z})$, out and in	edge pair → edge	Alg. 11, 12
6	$\mathbf{z} \leftarrow \mathbf{z} + \text{TriangleAttention}(\mathbf{z})$, start and end node	edge pair → edge	Alg. 13, 14
7	$\mathbf{z} \leftarrow \mathbf{z} + \text{MLP}(\mathbf{z})$	per edge	Alg. 15
Structure module, 8 blocks with shared weights. At the start $\mathbf{s}_u = \mathbf{m}_{1u}$ and T_u is the identity.			
8	$\mathbf{s} \leftarrow \mathbf{s} + \text{IPA}(\mathbf{s}, \mathbf{z}, T)$	node → node, values moved by $T_u^{-1} \circ T_v$	Alg. 22
9	$\mathbf{s} \leftarrow \mathbf{s} + \text{MLP}(\mathbf{s})$	per node	Alg. 20
10	$T_u \leftarrow T_u \circ \text{BackboneUpdate}(\mathbf{s}_u)$	per node, in its own frame	Alg. 23
Recycling, 4 cycles			
11	$\mathbf{z} \leftarrow \mathbf{z} + \text{Embed}(d_{uv})$, C$\beta$ distances from the frames	frames → edges	Alg. 32

Every line is $\mathbf{h}_u = \phi(\mathbf{x}_u, \bigoplus_v \psi)$ on nodes or on edges. Information moves in lines 1, 4, 5, 6, 8 and 11.

What exactly did each ablation remove?

SI 1.13.1 to 1.13.3

Row of Fig. 4a	What the model has instead
No raw MSA	the profile and the pairwise frequencies of the MSA, in one row, no column attention
No triangles, biasing or gating	axial attention on the pairs with a relative position bias, no gating
No IPA	frames from one linear projection of the single representation, no iteration
No recycling	a single pass through the network
No end-to-end structure gradients	the structure losses stop before the Evoformer, which learns from the auxiliary losses only
No templates	no template stack, four more MSA clusters instead

Baseline: the full model without self-distillation. Three or four seeds, not re-tuned

Not additive: no IPA costs 2.3, no recycling 4.5, both together 16.3 GDT

How is confidence predicted?

SI 1.9.6 and 1.9.7, Algorithm 29

pLDDT: a small network on the final single representation predicts, per residue, a distribution over 50 bins of lDDT-Cα. It is trained with cross-entropy against the lDDT-Cα of the prediction itself, on structures resolved between 0.1 and 3 Å

The reported pLDDT is the expected value of that distribution. The pLDDT of a chain is the mean over its residues

pTM, the predicted template modelling score: a pairwise head predicts the distribution of the aligned error of every residue pair, and the TM-score follows from these distributions

Confidence is a prediction, trained on the model's own errors

How was it trained?

Methods, “Training regimen”, SI 1.3 and 1.11.1, Table 4

Data: PDB structures released before 30 April 2018, and about 350,000 predicted structures of Uniclust30 sequences without a structure, filtered by confidence, 75 % of the samples

Stage 1: crops of 256 residues, about 10 million samples, about seven days on 128 TPU v3 cores

Stage 2: crops of 384 residues, the structural violation loss switched on, about 1.5 million samples, about four days

Losses: FAPE, torsion angles, distogram, masked MSA, pLDDT, experimentally resolved, violations

Eleven days on 128 TPU cores per model. Inference takes GPU minutes to hours.

AlphaFold 3: further changes

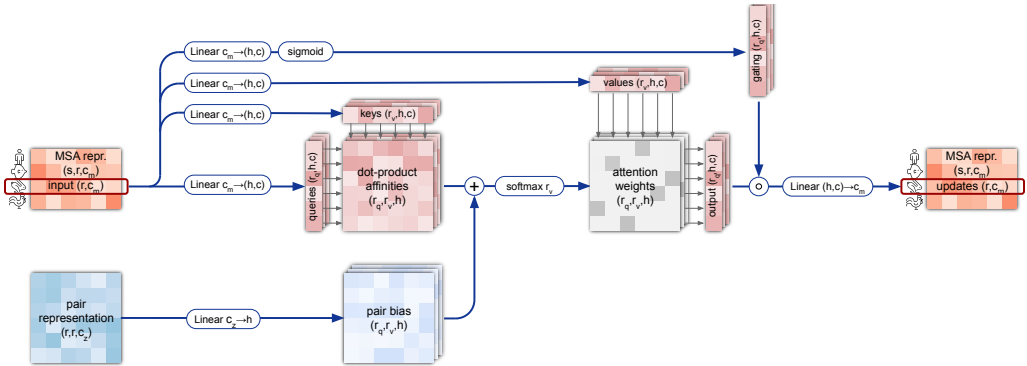
Abramson et al., *Nature* 630 (2024)

Input	tokens, one per residue or per ligand atom	p. 493
Trunk	pairformer, 48 blocks. MSA block of 4, MSA not kept	p. 494
Output	diffusion over raw atoms, no frames, random rotations instead	p. 495, SI Alg. 19
Confidence	trained on 20-step mini rollouts	p. 495
Sampling	5 seeds, 5 samples each, ranked with clash and chirality penalties	p. 494, 497
Limits	chirality 4.4 %, clashes, hallucination, no dynamics	p. 496, 497

Less structure in the network, more in the data and in the sampling

MSA row attention with pair bias

Supplementary Fig. 2, Algorithm 7



Supplementary Fig. 2. Every sequence of the MSA attends along itself, the pair representation biases the weights